IS 584: Deep Learning for Text Analytics

Project Proposal — Tip-of-the-Tongue Retrieval on TREC 2023

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***Abstract*—This proposal outlines a research project investigating the Tip-of-the-Tongue (ToT) known-item retrieval task using the TREC 2023 ToT track dataset. Tip-of-the-Tongue retrieval is a challenging problem characterized by high input noise: queries contain verbose, potentially misleading descriptions with little keyword overlap with the target item. I conduct a literature review of three related works, formulate research questions addressing noise reduction and model adaptation, and present exploratory data analysis of the dataset. My approach will leverage transformer-based architectures, specifically BERT-based dense retrievers, evaluated using NDCG@1000 and R@100 metrics.**

***Index Terms*—information retrieval, tip-of-the-tongue, TREC, dense retrieval, BERT, noise reduction**

I. Literature Review

A. *TREC 2023 Tip-of-the-Tongue Track Overview*

The overview paper of the TREC 2023 Tip-of-the-Tongue (ToT) track [1] establishes that the ToT task is significantly more difficult than classical retrieval tasks. The core challenge arises from query inputs that contain false or lacking information about the target item. The track attracted 11 search groups that submitted a total of 33 runs. Baseline systems included BM25, DistilBERT, and a GPT-4-based system. Model performance was evaluated using the NDCG@1000 metric, and the best-performing system achieved a score of 0.5554 on this metric. Several notable observations were reported: longer queries did not consistently improve retrieval performance; the use of external data was shown to improve results; and the median NDCG@1000 score across all submissions was 0, indicating that many models failed entirely on the task. The paper also highlights that horror and adventure movies were the most difficult categories to retrieve correctly.

B. *Generalizable Tip-of-the-Tongue Retrieval with LLM Re-ranking*

Prior methods in ToT retrieval were typically trained on domain-specific, small-scale datasets, limiting their generalizability [2]. To address this, the authors curated a new multi-domain dataset derived from Reddit, enabling broader coverage of ToT phenomena. The paper focuses on two key research contributions: (1) whether retrieval models can generalize across domains, and (2) whether large language models (LLMs) can serve as effective zero-shot re-rankers. The proposed pipeline operates in two stages. In the first stage, Dense Passage Retrieval (DPR) is employed as a first-stage retriever, using a BERT-based encoder to represent both queries and documents. In the second stage, GPT-4 is used as a zero-shot re-ranker: it receives only the query and a numbered list of candidate titles and reorders them by likelihood of being the correct answer. Key findings include: multi-domain training outperforms in-domain training on recall, GPT-4 is a strong zero-shot re-ranker, and a popularity bias was observed in GPT-4's re-ranking behavior.

C. *Webis at TREC 2023: Tip-of-the-Tongue Track*

The Webis submission [3] addresses a key limitation in ToT retrieval: long, verbose queries can mislead retrieval models by introducing noise. The paper proposes two query reduction approaches. First, the DeepCT model [4] is adapted to evaluate term importance in ToT queries. DeepCT is trained on the TOMT-KIS Reddit dataset such that terms appearing in the linked answer page receive high importance scores, while all other terms receive a score of 0. Second, GPT-3.5-turbo is used to remove unimportant details from queries through prompt-based reduction. Both approaches improve retrieval scores compared to the unprocessed query baseline. Notably, the ChatGPT-based reduction strategy outperforms the DeepCT-based approach, suggesting that semantic understanding of query relevance is more effective than term-level reweighting.

D. *Summary and Research Gap*

Collectively, these works demonstrate that ToT retrieval is a highly challenging task, with input noise being the primary obstacle. Each paper handles noise differently: Webis prunes or reweights it at the term level, while [2] decomposes the retrieval problem into a two-stage pipeline and uses GPT-4 to implicitly handle noisy descriptions. However, none of the reviewed works formally characterize which components of a ToT query carry signal versus noise across different domains. Moreover, the impact of domain-adapted tokenization on retrieval performance remains unexplored. These gaps motivate the research questions proposed in the following section.

II. Research Questions & Methodology

Based on the literature review, I formulate the following research questions:

- **RQ1:** Does changing the tokenizer affect retrieval performance on the Tip-of-the-Tongue task?
- **RQ2:** Can noise reduction techniques be further improved for retrieval performance?

A. Framework and Architecture

Experiments will be implemented using HuggingFace Transformers [5] and PyTorch. The primary model architecture will be BERT-based [6], as BERT is inherently designed as an encoder, making it well-suited for learning dense representations of both queries and documents. BERT will be used within a bi-encoder framework for first-stage retrieval.

B. Evaluation Metrics

Two evaluation metrics are selected, both commonly used in the ToT retrieval literature:

- **NDCG@1000**: Normalized Discounted Cumulative Gain at rank 1000. This is the primary metric used in the TREC 2023 ToT track [1], enabling direct comparison with reported results.
- **R@100**: Recall at rank 100. This measures the fraction of relevant documents retrieved within the top 100 results, capturing the model's ability to surface the correct item in an early candidate set, which is critical for downstream re-ranking stages.

C. Expected Outcomes

I expect that domain-adapted tokenization will improve the model's ability to handle rare or compound terms present in movie descriptions. Noise reduction is expected to benefit retrieval across all architectures.

III. Exploratory Data Analysis

The TREC 2023 ToT dataset consists of Wikipedia-based documents and movie-domain queries. Table I and Table III summarize key statistics.

A. Document Collection Statistics

TABLE I
Document Collection Key Statistics and Quality Checks

Metric	Value	Status
Total documents	231,852	✓
Duplicate document IDs	4	△
Docs with missing text	26	△
Docs with missing title	0	✓
Docs with 0 sections	9 (0.0%)	△
Docs with 0 infoboxes	9,766 (4.2%)	—

TABLE II
Document Field Distribution Statistics

Field	Min	Mean	Med	Max	Std
Text (words)	0	499.09	234	24,700	780.95
Text (chars)	0	2,983	1,404	149,049	4,705
Sections/doc	0	3.76	3	132	3.52
Infoboxes/doc	0	1.09	1	37	0.59

The document collection contains 231,852 entries with minor quality issues: 4 duplicate IDs, 26 documents with empty text fields, and 9 documents with no sections. Document text length follows a right-skewed distribution (mean: 499 words, median: 234 words), indicating a large number of short stub articles alongside a smaller number of very long documents.

B. Query Statistics

TABLE III
Query Length Distribution across Train and Dev Splits

Metric	Train (n=150)	Dev (n=150)
Total queries	150	150
Missing text	0	0
Query length (words)		
Min / Max	22 / 429	8 / 546
Mean / Median	138.85 / 119	131.20 / 111
Std	74.20	79.87
p25 / p75 / p95	93 / 173 / 286	78 / 166 / 285
Sentence annotations per query		
Min / Max	1 / 26	1 / 29
Mean / Median	8.37 / 7	7.33 / 6
Domain distribution	movie (100%)	movie (100%)

All queries in both splits belong to the movie domain. Queries are notably long: the mean query length is approximately 138 words in the training split and 131 words in the development split, with maximum lengths exceeding 400–500 words. This confirms the observation from [1] that ToT queries are verbose, and supports the motivation for noise reduction approaches. The absence of missing text in any query is a positive quality finding.

IV. Repository

A public Git repository has been initialized to track the project development. The repository follows the reproducible project template structure and currently includes the EDA scripts and preliminary data processing code used for this proposal.
Repository: https://github.com/atg93/project_584

AI Use Statement

AI-assisted grammar checking was applied to improve the clarity and correctness of written sentences throughout the proposal. Second, AI tools were used to convert the original content from a Word document (.docx) format into the IEEE conference LATEX template.

References

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